
Does Preprocessing Quality Matter for AutoML?

A Case Study on Time Series Forecasting with Environmental Data

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Abstract Although AutoML forecasting pipelines assume that input time series are complete and statistically consistent, real-world environmental monitoring data are often affected by missing values, sensor artefacts, and heterogeneous temporal dynamics. This paper reports on a case study that investigates whether domain-aware preprocessing significantly affects downstream AutoML forecasting performance using dissolved oxygen series from river basin monitoring stations. In order to do so, we propose a hierarchical data cleaning pipeline with distinct preprocessing operations. The proposed approach is evaluated using two AutoML forecasting frameworks, AutoGluon-TimeSeries and sktime, across four forecasting models and three preprocessing variants on six real-world stations. The data spans ten years of hourly measurements with rates of missing values up to 27%. Two types of experiments are performed, in which we assess: (i) the performance on the preprocessing task of imputing missing values and (ii) the performance on forecasting future values. Experimental results show that the hierarchical pipeline behaved better on six of seven evaluated metrics and reduced MASE by up to 69% for one of the models we experimented with (DeepAR) in stations with a high number of missing values. Most importantly, this work provides a new real-world dataset based on dissolved oxygen environmental data, which highlights the importance of practical issues, such as implausible values and ranges of missing values, and shows the performance gains that AutoML can achieve on this particular real-world task.

1 Introduction

Environmental water monitoring networks generate continuous streams of water quality measurements that support operational and regulatory decisions in river basins and reservoirs (Chapra 2008). Dissolved oxygen (DO) is particularly relevant (D. Li et al. 2022), as it reflects the combined influence of physical transport, biological productivity, and organic matter decomposition, making it a sensitive indicator of ecosystem condition and anthropogenic pressure. Reliable forecasting of dissolved oxygen concentrations is thus important for pollution monitoring, infrastructure management, and early warning systems.

Despite this practical importance, time series collected from automated field sensors represent a challenging domain for machine learning (ML) applications. Monitoring stations are often affected by sensor drift, fouling, communication faults, and transmission errors (Rode et al. 2016), resulting in missing observations, physically implausible values, and prolonged gaps without reliable measurements. In long-term high-frequency datasets (e.g., hourly measures), even moderate missing rates can substantially distort the temporal structure of the series and introduce biases that propagate into downstream forecasting models (Buuren 2018).

This challenge is particularly acute in Automated Machine Learning (AutoML) settings (He et al. 2021), where pipelines automate model selection, hyperparameter optimisation, and ensemble

construction under the implicit assumption that input data are complete and statistically consistent. When this assumption is violated, preprocessing artefacts are not isolated errors: they propagate through every subsequent modelling decision, potentially degrading forecast accuracy and producing physically implausible predictions (Hyndman et al. 2018). Thus, preprocessing becomes a critical architectural decision rather than a routine data cleaning step.

From a data-centric artificial intelligence (AI) perspective (Zha et al. 2025), preprocessing is not merely a preliminary operation but part of the learning system itself, since the statistical properties induced during important preprocessing operations, such as imputation, directly influence model selection, optimisation, and ensemble behaviour. Sambasivan et al. 2021 specifically research data quality, and coin the term *data cascades*, i.e., compounding downstream failures caused by seemingly minor issues in training data. Through interviews with 53 AI practitioners, the authors show that data work is systematically undervalued and deprioritised in favour of model development, and argue that rigorous data preprocessing and quality practices are essential to prevent these cascading failures. Although substantial effort has been devoted to missing value imputation and time series forecasting, these problems are commonly treated in isolation (Khayati, Lerner, et al. 2020). Imputation methods are frequently benchmarked under controlled, idealised conditions that do not reflect the heterogeneous dynamics of operational environmental monitoring. At the same time, most AutoML benchmarking studies assume complete or near-complete input data (Shchur et al. 2023; Mischos et al. 2025), while studies that explicitly address missing data either evaluate imputation independently from forecasting or adopt end-to-end architectures designed to bypass preprocessing entirely (P.-C. Li et al. 2026). The joint effect of preprocessing quality on AutoML forecasting performance therefore remains insufficiently understood, particularly in real-world scenarios characterized by *heterogeneous temporal regimes* (i.e., the time series does not show a clear seasonality or obvious pattern) and high amounts of missing values.

To address this gap, we present a case study on dissolved oxygen forecasting from real-world river basin monitoring stations that combines several data cleaning operators that are inspired by domain knowledge. Some important characteristics of the dissolved oxygen case study are (i) multi-model time series data that provides summary statistics of each hour’s measurements (e.g., min, max, mean, median, etc.), (ii) measurements across various stations with different data characteristics, and (iii) larger ranges of faulty or missing sensor measurements that need to be identified and imputed. We utilise a pipeline consisting of various data cleaning steps, and evaluate it using AutoGluon-TimeSeries (Shchur et al. 2023) and sktime (Löning et al. 2019) over a 91-day prediction horizon on six stations spanning ten years of hourly measurements. The main contributions of this work are: (i) we provide a new real-world river basin environmental monitoring dataset, characterised by heterogeneous temporal regimes and rates of missing values up to 27%; (ii) a data-centric analysis of how preprocessing quality affects AutoML forecasting behaviour under challenging missing value conditions (iii) a comparative evaluation of preprocessing sensitivity across two AutoML forecasting frameworks and four forecasting architectures; and (iv) empirical evidence that AutoML delivers substantial gains on this type of real-world environmental monitoring task once input data meet a minimum threshold of temporal coherence.

2 Related Work

AutoML frameworks have demonstrated strong performance across a wide range of forecasting tasks (Baratchi et al. 2024), yet most benchmarking studies assume complete or near-complete input data, as noted by Mischos et al. (2025). This assumption rarely holds in operational environmental monitoring, where sensor drift, communication faults, and prolonged outages routinely produce heterogeneous patterns of missing values that violate the assumptions about the missing-data process made by automated pipelines, as documented by Rode et al. (2016). This study examines the performance of AutoML methods in the context of faulty measurements and missing values for time series.

From a data-centric AI perspective (Zha et al. 2025), data quality is not a preliminary concern but an architectural one: preprocessing decisions directly shape the statistical properties seen by model selection, hyperparameter optimisation, and ensemble construction. Whang et al. (2023) show that data quality improvements can produce gains comparable to or larger than increases in model complexity, a finding particularly relevant in AutoML settings where upstream artefacts propagate throughout the pipeline (Hyndman et al. 2018). Missing data remain one of the principal challenges in this regard, motivating imputation strategies ranging from classical statistical to deep learning approaches (Whang et al. 2023), with evidence that no single method dominates across all gap sizes, and that imputation choice has measurable downstream effects on forecasting accuracy, as shown by Khayati, Lerner, et al. (2020).

The importance of domain context in preprocessing has been demonstrated in environmental monitoring settings. Passarini et al. (2022) show that incorporating geographic and station-level characteristics into imputation improves reconstruction accuracy relative to generic procedures, reinforcing the view that domain-specific characteristics must inform pipeline design rather than be treated as incidental properties of the data.

In water quality forecasting specifically, O’Donncha et al. (2022) benchmark a bidirectional LSTM against AutoML-generated baselines for dissolved oxygen prediction at an aquaculture site, and compare it against AutoML methods (specifically IBM Watson Studio’s AutoAI module and AutoMLPipeline (Palmes et al. 2021)). We extend this line of work to river basin monitoring under substantially higher rates of missing values, and shift the focus from architecture selection to the question of how preprocessing quality propagates into AutoML forecasting behaviour across multiple frameworks and model families. Additionally, by using AutoGluon and sktime, we utilise AutoML frameworks that are more common in the AutoML community.

Automated method selection for imputation has itself become a research direction, as exemplified by Khayati, Chacun, et al. (2025), reflecting a broader move toward robust performance under realistic data conditions. The dataset introduced here contributes to this agenda by providing a real-world benchmark characterised by heterogeneous temporal regimes and substantial amounts of missing values, where the interaction between preprocessing and AutoML pipeline behaviour can be studied under conditions that reflect practical deployment constraints rather than controlled experimental settings.

3 Dissolved Oxygen Time-series

Dissolved oxygen time series were obtained from the Integrated Water Quality Monitoring System (SIMQUA) (CETESB 2026) of the Environmental Company of the State of São Paulo (CETESB). The dataset comprises six monitoring stations across the Tietê River Basin, São Paulo, Brazil (Figure 1), spanning January 2015 to April 2025 at hourly resolution (89,832 steps per station). For each hour, the system records the minimum, maximum, mean, median, standard deviation, and variance of the dissolved oxygen measurements. In this study, we focus on predicting the hourly mean dissolved oxygen concentration. We make all raw series, quality-control outputs, and gap-filled variants publicly available, together with the notebooks required to reproduce the full preprocessing and forecasting experiments.¹

The stations generally record dissolved oxygen measurements on an hourly basis. However, missing values occur at varying intervals, primarily due to sensor errors, communication failures, or routine maintenance. Table 1 summarizes the intervals of missing data. The amount of missing data ranges from 6.51% (EF07) to 27.00% (EF03). In total, we identified 914 gaps across all stations. Gap-length distribution varies substantially across stations: short gaps (≤ 10 h) predominate at EF01, EF06, EF07, and EF08, while EF02 is the most challenging station, with 52.5% of its gaps exceeding 72 h.

¹Full details: <https://github.com/dallasta-gui/Preprocessing-for-AutoML>

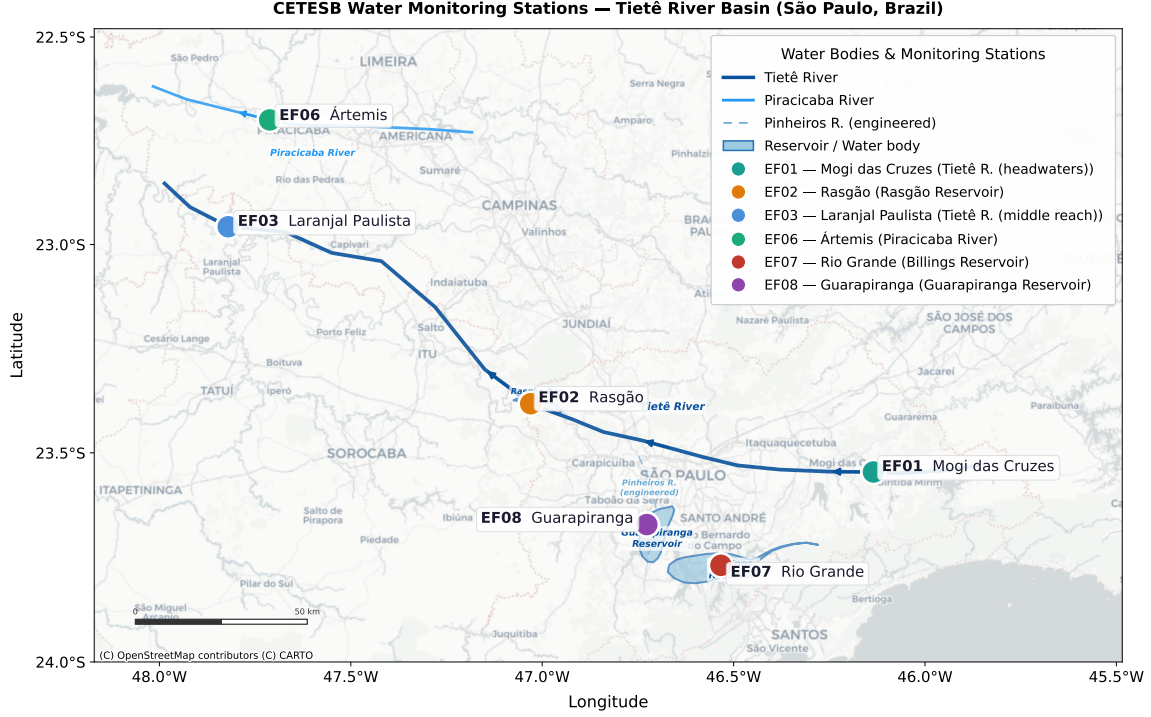


Figure 1: CETESB monitoring stations across the Tietê River Basin. Stations span river reaches and reservoirs, yielding heterogeneous dissolved oxygen dynamics.

Table 1: Summary of dissolved oxygen datasets and gap-length distribution per station. Total Missing (%) is computed over the total series length (89,832 hourly steps). Short: ≤ 10 h; Medium: 11–72 h; Long: > 72 h. Values in the last four columns represent the number of gaps.

Station	Type	DO Regime	Total Missing (%)	Short (≤ 10 h)	Medium (11–72 h)	Long (> 72 h)	Total Gaps
EF01	River	Intermediate	8.38	80	20	20	120
EF02	Reservoir	Hypoxic-impacted	22.79	37	84	134	255
EF03	River	Intermediate	27.00	71	46	34	151
EF06	River	Diel-productive	17.57	86	18	23	127
EF07	Reservoir	Intermediate	6.51	37	6	13	56
EF08	Reservoir	Diel-productive	26.21	137	51	17	205
Total				448	225	241	914

We identified different behaviour at different stations (due to the characteristics and heterogeneity of the various sites). We assigned each station a behavioral regime based on five summary statistics computed from its quality-controlled DO series: standard deviation, median daily amplitude (amp_dia, median of daily max – min), lag-24 autocorrelation (acf_24, capture the persistence of the diel cycle), fraction of readings below the hypoxic threshold of 2 mg/L (p_low), and the strength of the average diel cycle (diel_strength, the range of the hour mean DO). Stations were classified using the ordered rules as stated in Figure 2.

The six stations for which we have data all fall in either of the following categories: Intermediate, Diel-productive, and Hypoxic-impacted.

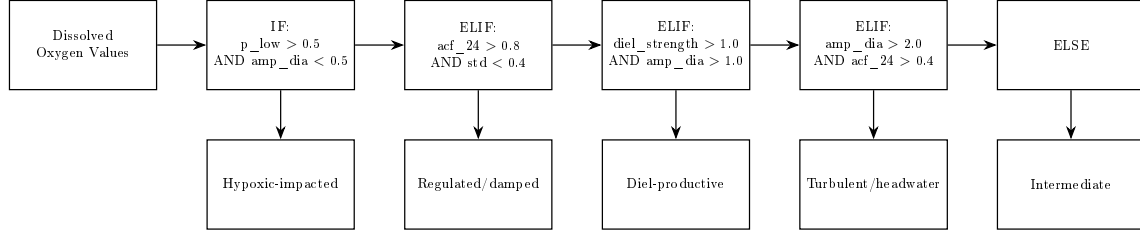


Figure 2: Ordered decision rules used to assign each station a behavioral regime from its five summary statistics (std , amp_dia , acf_24 , p_low , and diel_strength ; defined in the text). The rules are evaluated from top to bottom: a station is assigned the regime of the first condition its statistics satisfy, *Hypoxic-impacted*, *Regulated/damped*, *Diel-productive*, or *Turbulent/headwater*, and any station matching none of the conditions is labeled *Intermediate*.

4 Data Cleaning Pipeline inspired by Domain Expertise

The preprocessing and forecasting pipeline employed in this case study is informed by domain expertise and described in the following sections. Figure 3 summarises this pipeline.

4.1 Domain-knowledge Inspired Quality Control

Raw dissolved oxygen series undergo two quality-control steps before imputation. Note that these steps potentially remove additional values to be imputed. First, a *physical plausibility filter* removes observations outside $[0, 14.5]$ mg/L, the physically admissible range for dissolved oxygen.

Second, we aim to detect regions of (almost) constant values, caused by faulty sensor measurements or drift, using a rolling window evaluated against three joint criteria: the observed range ($\text{max} - \text{min}$), the standard deviation, and the magnitude of a linear regression slope fitted within the window. For each type of station, we determined specific thresholds, and a given window of measurements is flagged as an artefact when all three quantities fall simultaneously below regime-dependent thresholds (Table 6, appendix). We remove those data from the series (meaning that they need to be imputed afterwards). Of course, one of the challenges involves ensuring that genuinely flat, low-variance, and non-trending segments are distinguished from sensor failures within an otherwise dynamic signal. The threshold is calibrated per station according to the behavioural regime assigned in Section 3, ensuring that flat-line detection remains sensitive across stations with qualitatively distinct dynamics. The thresholds are set in consultation with the domain expert.

4.2 Imputation using the Hierarchical Gap-Filling Strategy

We experiment with five imputation strategies, ranging from simple local methods to global model-based approaches: *forward filling* (Moritz et al. 2017) (replace with last known value, only used as baseline), *linear interpolation* (Moritz et al. 2017) (linear transition between boundary observations, both baseline and in the pipeline), *moving average* (Moritz et al. 2017) (mean of a

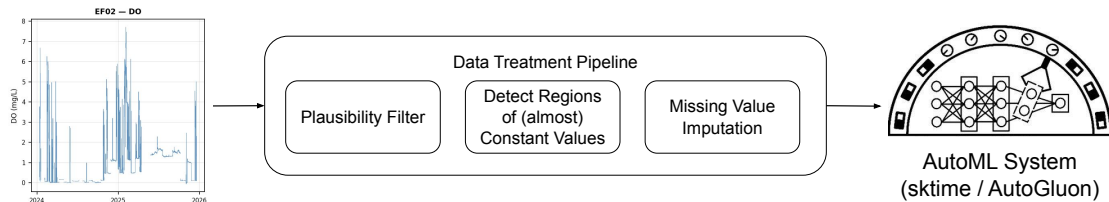


Figure 3: The domain-expertise-inspired pipeline that we utilise on the dissolved oxygen time series.

Table 2: Gap-length boundaries for the hierarchical imputation strategy. The **Tier** column maps each boundary rule onto the Short/Medium/Long classification used elsewhere in the paper (Tables 1 and 3); the Medium tier is served by two distinct reconstruction rules depending on sub-range.

Boundary	Range (h)	Tier	Imp. method	Rationale
$t \leq T_1$	≤ 10	Short	Interpolation	Linear transition remains physically plausible over short gaps.
$T_1 < t \leq T_2$	11–42	Medium	STL	Diel seasonality becomes evident; decomposition reconstructs the underlying cycle.
$T_2 < t \leq T_3$	43–72	Medium	Moving Average	Longer gaps benefit from local smoothing while avoiding strong trend assumptions.
$t > T_3$	> 72	Long	Prophet	Long gaps require a global seasonal model to generate plausible estimates.

symmetric local window of half-size w), *Seasonal and Trend decomposition using Loess (STL)-based imputation* (Cleveland et al. 1990) (reconstruct from estimated trend and seasonal components), and *Prophet* (Taylor et al. 2018) (decomposable additive model fitted on available observations). We determined various tiers of gap lengths, ranging from short to long. We select the imputation strategy based on the length. These are hierarchically ordered by length. Forward filling is not used, as it imputes segments with a constant value that would be indistinguishable from the constancy artefacts removed during quality control. Table 2 shows the imputation strategy per gap length.

The thresholds T_1 , T_2 , and T_3 reflect domain judgment, developed in consultation with a domain expert in water quality monitoring: $T_1 = 10$ h separates sub-diel gaps, where linear interpolation remains physically plausible; $T_2 = 42$ h marks the transition from seasonal-trend (STL) reconstruction to a centred moving average; and $T_3 = 72$ h corresponds to three full diel cycles, beyond which even a local moving average loses the reference required for reconstruction and a global seasonal model is required.

4.3 Forecasting Frameworks and Models

After the time series have been treated for data quality and the missing values have been imputed, we can utilise time series prediction to model the time series. We evaluate two AutoML frameworks on this task: **AutoGluon-TimeSeries** (Shchur et al. 2023), which automates training and ensembling within a unified interface, and **sktime** (Löning et al. 2019), a modular library that supports AutoML workflows by enabling systematic construction, comparison, and automated selection of forecasting pipelines through a unified scikit-learn interface. Both are evaluated under the same set of four models: DeepAR (Salinas et al. 2020) (autoregressive probabilistic recurrent neural network, RNN), XGBoost (Chen et al. 2016) (direct tabular forecasting), Seasonal Autoregressive Integrated Moving Average (SARIMA (Box et al. 2015), a classical statistical model with seasonal components), and Prophet (Taylor et al. 2018) (decomposable additive model). Using a common model set ensures that performance differences are attributable to pipeline design rather than model availability.

5 Experimental Setup

We evaluate two specific tasks in isolation: preprocessing quality and forecasting performance.

Preprocessing evaluation compares the Hierarchical Pipeline against two baselines: Baseline I (forward fill + backward fill) and Baseline II (linear interpolation + boundary fill). We evaluate the quality via (i) a holdout protocol with 100 artificially generated gaps, stratified by gap-length tier, using Dynamic Time Warping (DTW) distance constrained by a Sakoe-Chiba band (Sakoe et al. 1978), mean bias, and Kolmogorov-Smirnov (KS) statistic; and (ii) an expanding-window time

series cross-validation (tsCV) (Hyndman et al. 2018) starting from one year, expanding by one quarter per fold, evaluated at three representative gap sizes. Prophet is excluded from tsCV due to computational cost.

Forecasting evaluation uses a fixed temporal split: training from January 2015 to December 2024, test from January to March 2025 (2,184 hours, 91 days). Each framework is evaluated on three preprocessing variants: Baseline I, Baseline II, and the Hierarchical Pipeline. Models are configured as follows: DeepAR with context window 168 h, 2 layers, hidden size 64, 50 epochs; XGBoost with 300 estimators and learning rate 0.05; SARIMA with auto-selected orders; Prophet with default structure. Performance is measured using MASE scaled to a 24-hour seasonal period.

6 Experimental Results

6.1 Preprocessing Quality

The main results are shown in Figure 4 and Table 7 (appendix). The Hierarchical Pipeline consistently outperforms both baselines across six of seven reconstruction metrics. We observe the largest improvement in distributional fidelity, i.e., the extent to which the reconstructed values preserve the statistical distribution of the original time series. KS statistic drops from 0.74 under Baseline I to 0.46 under the Hierarchical Pipeline, a 38% reduction, indicating that the reconstructed series more closely matches the empirical distribution of the original signal rather than merely minimising pointwise error. Directional accuracy (DA) rises from near-zero under Baseline I to 0.68, reflecting the near-complete elimination of flat-line artefacts that suppress temporal variation. The Mean Absolute Error of Differences (MAED) is an exception, showing no meaningful difference across methods. This suggests that preserving changes between consecutive observations is less sensitive to the choice of imputation strategy than preserving broader temporal structure and the overall data distribution.

Table 3 sorts these results by gap size. This gap-size stratification reveals where the Hierarchical Pipeline adds the most value. For short gaps (≤ 10 h), performance is nearly identical to Baseline II (DTW: 0.821 vs. 0.816), consistent with the design: both use linear interpolation at this tier. For medium gaps (42 h), DTW falls from 5.220 under Baseline I to 2.337, a 55% reduction. For long gaps (72 h), the reduction is 35% relative to Baseline I. The gains grow with gap length precisely because longer gaps require methods capable of capturing seasonal cycles and long-term trends, which purely local methods cannot provide. The domain expert verified that the reconstruction quality

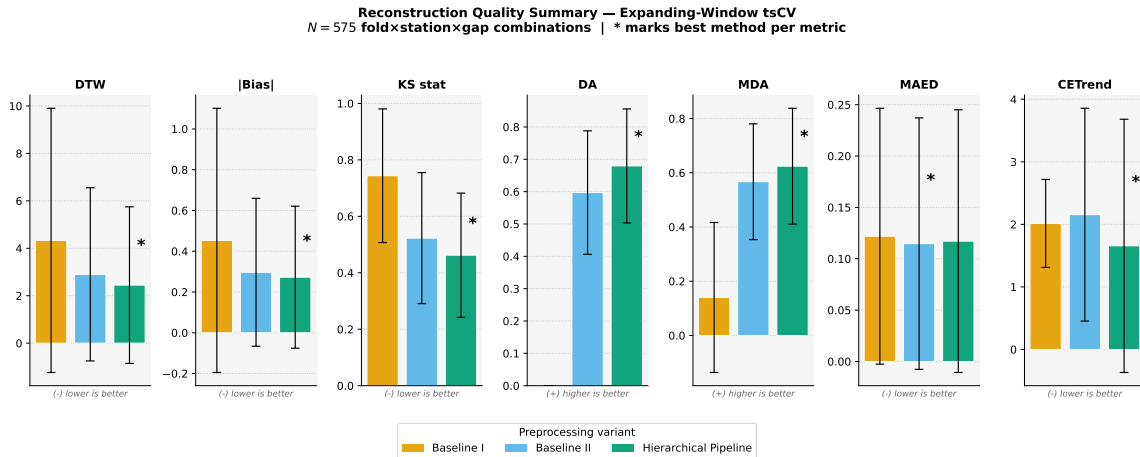


Figure 4: Reconstruction quality metrics aggregated over $N = 575$ fold-station-gap combinations. Error bars represent ± 1 standard deviation calculated across the 575 time-series cross-validation combinations. Asterisk marks best method per metric.

Table 3: Mean DTW by gap size (standard deviations calculated over the average DTW per gap). Gains grow with gap length, reflecting the increasing inadequacy of local methods as temporal complexity increases.

Gap size (h)	Baseline I	Baseline II	Hierarchical
Short (≤ 10)	1.415 ± 1.866	0.816 ± 1.052	0.821 ± 1.053
Medium (11 – 72)	5.220 ± 5.563	3.479 ± 3.541	2.337 ± 2.785
Long (> 72)	6.828 ± 6.822	4.741 ± 4.438	4.456 ± 4.302

remains stable across all tsCV folds with no systematic degradation over the ten-year evaluation period (see also Figure 9, appendix), confirming that the pipeline performs robustly to hydrological variability through the year.

Figure 5 shows the results of the full preprocessing pipeline applied to the observed data on two stations. See Figure 6 (appendix) for all six stations.

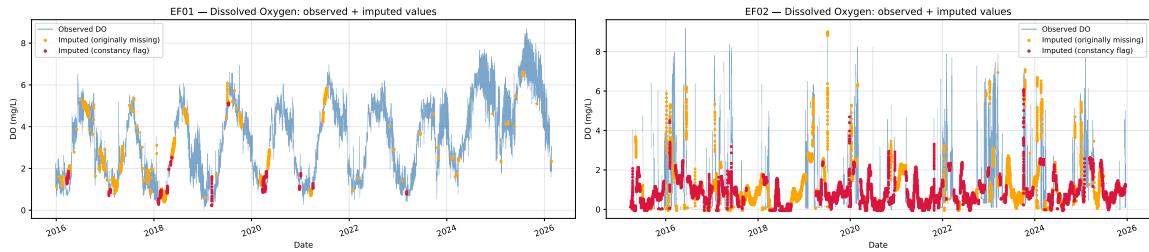


Figure 5: Dissolved oxygen at stations EF01 and EF02 after quality control and hierarchical gap-filling. Blue lines represent observed values retained after quality control. Orange points indicate positions reconstructed due to originally missing observations. Red points indicate positions where constancy detection flagged extended flat-line segments as probable sensor artefacts; these observations were removed during quality control and subsequently reconstructed by the imputation pipeline.

6.2 Forecasting Performance

The following results examine how preprocessing quality propagates into forecasting performance across frameworks and models. Station-level MASE summaries are reported in Tables 4 and 5; per-model details are in Tables 8 and 9 (appendix).

Using the series treated by our pipeline, **AutoGluon-TimeSeries** outperformed the baseline preprocessing approaches at five of the six stations. The largest gains occurred for DeepAR, which reduced MASE by 69% at EF01 ($19.71 \rightarrow 6.11$), 54% at EF06 ($13.92 \rightarrow 6.39$), and 51% at EF07 ($18.13 \rightarrow 8.81$), leading to qualitatively better forecasts at the stations with the highest rates of missing values and the strongest temporal structure (e.g., persistent diel cycles and temporal autocorrelation). SARIMA performance also improved, with MASE reductions of 58% at EF06 and 40% at EF02. At the aggregate level, EF06 and EF07 had reductions of 47% and 38% across all four models. The exception was EF03, with a marginal mean increase of 1%, consistent with its already low baseline MASE and intermediate behavioural regime: when a series is already well-behaved, and gaps are infrequent, preprocessing quality contributes little additional gain.

sktime showed a mixed but overall positive response to the Hierarchical Pipeline. Forecast accuracy improved at four of the six stations, with all four forecasting models achieving lower MASE at EF02, EF03, and EF07. The main exception was EF06, where the aggregate MASE increased by 72%, driven entirely by SARIMA ($21.76 \rightarrow 42.29$, +94%). This reflects a model-specific limitation

Table 4: MASE summary per station — AutoGluon-TimeSeries. Δ_I and Δ_{II} : percentage change from Baseline I and Baseline II to the Hierarchical Pipeline, respectively ($\downarrow$ improvement, $\uparrow$ degradation). *Signal* reports, out of the four forecasting models (DeepAR, Prophet, SARIMA, XGBoost; per-model values in Table 8, appendix), how many improved ($\downarrow$) or degraded ($\uparrow$) under the Hierarchical Pipeline, where each model’s direction is determined by whichever of Δ_I or Δ_{II} has the larger absolute magnitude.

Station	BI	BII	Hierarchical	Δ_I	Δ_{II}	Signal
EF01	18.79	18.90	15.41	−18.00%	−18.47%	3 $\downarrow$ 1 $\uparrow$
EF02	9.52	8.10	7.54	−20.80%	−11.23%	3 $\downarrow$ 1 $\uparrow$
EF03	6.00	5.89	6.07	+1.04%	+4.17%	2 $\downarrow$ 2 $\uparrow$
EF06	7.70	7.12	4.08	−47.01%	−24.68%	3 $\downarrow$ 1 $\uparrow$
EF07	6.54	6.33	4.08	−37.61%	−35.55%	4 $\downarrow$ 0 $\uparrow$
EF08	2.32	2.32	2.05	−11.84%	−11.37%	2 $\downarrow$ 2 $\uparrow$

rather than a general failure of the preprocessing strategy. To achieve stationarity, SARIMA applies first-order and seasonal differencing. At EF06, these transformations amplified the strongly preserved diel cycle in the reconstructed series, causing the model to extrapolate a persistent trend over the 91-day forecast horizon and producing predictions that rapidly exceeded physically plausible dissolved oxygen values. The other three models at EF06 (DeepAR, Prophet, XGBoost) either improved or remained stable.

Table 5: MASE summary per station — sktime. Δ_I and Δ_{II} : percentage change from Baseline I and Baseline II to the Hierarchical Pipeline, respectively ($\downarrow$ improvement, $\uparrow$ degradation). *Signal* reports, out of the four forecasting models (DeepAR, Prophet, SARIMA, XGBoost; per-model values in Table 9, appendix), how many improved ($\downarrow$) or degraded ($\uparrow$) under the Hierarchical Pipeline, where each model’s direction is determined by whichever of Δ_I or Δ_{II} has the larger absolute magnitude.

Station	BI	BII	Hierarchical	Δ_I	Δ_{II}	Signal
EF01	20.08	20.18	19.83	−1.27%	−1.73%	3 $\downarrow$ 1 $\uparrow$
EF02	13.14	11.11	8.61	−34.47%	−22.50%	4 $\downarrow$ 0 $\uparrow$
EF03	4.74	4.95	4.37	−7.81%	−11.76%	4 $\downarrow$ 0 $\uparrow$
EF06	7.22	7.48	12.44	+72.30%	+66.31%	1 $\downarrow$ 3 $\uparrow$
EF07	3.51	3.61	3.38	−3.56%	−6.37%	4 $\downarrow$ 0 $\uparrow$
EF08	8.05	8.32	6.32	−21.46%	−24.04%	3 $\downarrow$ 1 $\uparrow$

Wilcoxon signed-rank tests were computed per metric across all model–station blocks ($n = 24$), comparing Baseline I, Baseline II, and the Hierarchical Pipeline pairwise (Figure 10, appendix). For AutoGluon-TimeSeries, the Hierarchical Pipeline differed significantly ($\alpha = 0.05$) from both baselines in DA, MDA, MAED and CETrend, while for sktime, significant differences were concentrated between the two baselines themselves rather than between either baseline and the Pipeline. To complement these pairwise tests, we additionally applied a Friedman test followed by Nemenyi post-hoc analysis over the same model–station pairs ($n = 24$), as shown in Figures 11 (appendix). The Friedman test indicated statistical significance for both frameworks, and the Hierarchical Pipeline achieved the best average rank in both cases. For AutoGluon-TimeSeries, the Hierarchical Pipeline was significantly better than Baseline I but not from Baseline II, while for sktime none of the three pipelines differed significantly from each other.

6.3 Limitations

This work provided a valuable case study of AutoML methods on time series of dissolved oxygen. The evaluation covered six stations from a single river basin in southeastern Brazil. The three

behavioural regimes identified here, intermediate, diel-productive, and hypoxic-impacted, may not exhaustively characterise the diversity of dissolved oxygen dynamics encountered in other basins, climatic zones, or aquatic system types. The generalizability of the forecasting sensitivity patterns documented in this case study remains to be established in broader geographic and hydrological contexts.

7 Discussion and Future Work

We introduce a real-world time series dataset from the environmental monitoring domain and examine what practical challenges AutoML must address to be effective in this setting, most prominently, the identification of faulty measurements and imputation of gaps ranging from short dropouts to multi-week gaps.

The Dataset as a Benchmark Contribution The six-station Tietê River Basin dataset introduced here provides the AutoML community with a real-world dissolved oxygen benchmark explicitly characterised by heterogeneous profiles of missing values, spanning a decade of hourly measurements with rates up to 27% and a gap-length distribution that ranges from sub-hourly dropouts to multi-week outages. It reflects a realistic data science scenario of environmental monitoring networks, where sensor drift, faulty measurements, and communication faults produce data quality problems that are structurally different from the controlled, uniformly-distributed gaps used in most imputation benchmarks (Khayati, Lerner, et al. 2020).

AutoML Gains on Real-World Environmental Data Taken together, the results provide empirical evidence that AutoML frameworks can deliver meaningful performance gains on real-world environmental monitoring tasks, but in this specific case study the input data required meet a quality standard. We speculate that the performance gains were mostly obtained through the data preprocessing steps of the pipeline, which made the data suitable for time series forecasting.

Future Work Beyond benchmarking practice, our work points to a specific opportunity for AutoML system design: None of the AutoML frameworks allowed for inserting the domain expertise that was required for treating the time series with the specifics that are important for dissolved oxygen measurements (i.e., the large gaps that require a different treatment per station type), while this domain is especially a nice use case for AutoML tools. Future work should focus on extending the capabilities of AutoML systems to better cater to the needs of specific application domains.

Focussing more on the specific domain application, future work could extend the evaluation to additional water quality variables, multiple river basins, and longer forecast horizons, where the propagation of preprocessing artefacts into AutoML behaviour is expected to be more severe. More broadly, the case study presented here points to the need for AutoML benchmarks that explicitly characterise the structure of missing values, not just overall missing rate, and that evaluate preprocessing sensitivity as a first-class experimental dimension alongside model architecture and ensemble design. This points toward AutoML systems that treat reconstruction-strategy selection as a first-class, jointly-optimised component of the pipeline, for example, meta-learned selection of imputation method conditioned on detected gap length and regime, rather than a fixed preprocessing stage decided independently of the AutoML search.

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References

- Baratchi, Mitra et al. (2024). “Automated machine learning: past, present and future”. In: *Artificial Intelligence Review* 57 (5).
- Box, George E. P. et al. (2015). *Time Series Analysis: Forecasting and Control*. 5th. Hoboken, NJ: John Wiley & Sons.
- Buuren, Stef van (2018). *Flexible Imputation of Missing Data*. CRC Press. URL: <https://stefvanbuuren.name/fimd/>.
- CETESB (2026). *SIMQUA – Sistema de Monitoramento da Qualidade das Águas*. Accessed: 2026-05-09. URL: <https://simqua.cetesb.sp.gov.br/>.
- Chapra, Steven C. (2008). *Surface Water-Quality Modeling*. Waveland Press.
- Chen, Tianqi and Carlos Guestrin (2016). “XGBoost: A Scalable Tree Boosting System”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, pp. 785–794.
- Cleveland, Robert B. et al. (1990). “STL: A Seasonal-Trend Decomposition Procedure Based on Loess”. In: *Journal of Official Statistics* 6.1, pp. 3–73.
- He, Xin, Kaiyong Zhao, and Xiaowen Chu (2021). “AutoML: A Survey of the State-of-the-Art”. In: *Knowledge-Based Systems* 212, p. 106622.
- Hyndman, Rob J. and George Athanasopoulos (2018). *Forecasting: Principles and Practice*. 2nd. OTexts.
- Khayati, Mourad, Guillaume Chacun, et al. (2025). “A-DARTS: Stable Model Selection for Data Repair in Time Series”. In: *Proceedings of the IEEE 41st International Conference on Data Engineering (ICDE)*, pp. 2009–2023.
- Khayati, Mourad, Alberto Lerner, et al. (2020). “Mind the Gap: An Experimental Evaluation of Imputation of Missing Values Techniques in Time Series”. In: *Proceedings of the VLDB Endowment* 13.5, pp. 768–782.
- Li, Daoliang, Mi Zou, and Lingwei Jiang (2022). “Dissolved Oxygen Control Strategies for Water Treatment: A Review”. In: *Water Science & Technology* 86.6, pp. 1444–1466.
- Li, Peng-Cheng et al. (2026). “MeaTS: An End-to-End Meta-Enhanced Attention for Time-Series Forecasting With Missing Data”. In: *IEEE Transactions on Industrial Informatics* 22.5.
- Löning, Markus et al. (2019). *sktime: A Unified Interface for Machine Learning with Time Series*. Presented at the NeurIPS 2019 Workshop on Systems for ML. arXiv: 1909.07872 [cs.LG]. URL: <https://arxiv.org/abs/1909.07872>.
- Mischos, Stavros et al. (2025). “Energy Time Series Forecasting: A Modular and Automated Approach for Benchmarking”. In: *2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*.
- Moritz, Steffen and Thomas Bartz-Beielstein (2017). “imputeTS: Time Series Missing Value Imputation in R”. In: *The R Journal* 9.1, pp. 207–218.
- O’Donncha, Fearghal et al. (2022). “A spatio-temporal LSTM Model to Forecast Across Multiple Temporal and Spatial Scales”. In: *Ecological Informatics* 69, p. 101687.
- Palmes, Paulito P. et al. (2021). “Designing Machine Learning Pipeline Toolkit for AutoML Surrogate Modeling Optimization”. In: *CoRR* abs/2107.01253. arXiv: 2107.01253.
- Passarini, Pablo de O. et al. (2022). “Sistema de Geração de Dados Faltantes em Séries Temporais Climatológicas usando Triangulação, Aprendizado e Meta-Aprendizado de Máquinas”. In: *Proceedings of the XXIII Brazilian Symposium on Geoinformatics (GEOINFO)*.
- Rode, Michael et al. (2016). “Sensors in the Stream: The High-Frequency Wave of the Present”. In: *Environmental Science & Technology* 50.19, pp. 10297–10307.

- Sakoe, Hiroaki and Seibi Chiba (1978). "Dynamic Programming Algorithm Optimization for Spoken Word Recognition". In: *IEEE Transactions on Acoustics, Speech, and Signal Processing* 26.1, pp. 43–49.
- Salinas, David et al. (2020). "DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks". In: *International Journal of Forecasting* 36.3, pp. 1181–1191.
- Sambasivan, Nithya et al. (2021). "Everyone wants to do the model work, not the data work": Data Cascades in High-Stakes AI". In: *CHI '21: CHI Conference on Human Factors in Computing Systems, Virtual Event / Yokohama, Japan, May 8-13, 2021*. ACM, 39:1–39:15.
- Shchur, Oleksandr et al. (2023). "AutoGluon-TimeSeries: AutoML for Probabilistic Time Series Forecasting". In: *Proceedings of the Second International Conference on Automated Machine Learning*. Vol. 224. Proceedings of Machine Learning Research. PMLR, pp. 9/1–21.
- Taylor, Sean J. and Benjamin Letham (2018). "Forecasting at Scale". In: *The American Statistician* 72.1, pp. 37–45.
- Whang, Steven Euijong et al. (2023). "Data Collection and Quality Challenges in Deep Learning: A Data-Centric AI Perspective". In: *The VLDB Journal* 32.4, pp. 791–813.
- Zha, Daochen et al. (2025). "Data-Centric Artificial Intelligence: A Survey". In: *ACM Computing Surveys* 57.5, pp. 1–42.

Submission Checklist

1. For all authors...

- (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]
- (b) Did you describe the limitations of your work? [Yes] Limitations are discussed in Section 6.3.
- (c) Did you discuss any potential negative societal impacts of your work? [No] The work focuses on environmental water quality monitoring, a domain with broadly positive societal impact. No negative societal impacts were identified.
- (d) Did you read the ethics review guidelines and ensure that your paper conforms to them? (see <https://2022.automl.cc/ethics-accessibility/>) [Yes]

2. If you ran experiments...

- (a) Did you use the same evaluation protocol for all methods being compared (e.g., same benchmarks, data (sub)sets, available resources, etc.)? [Yes] All preprocessing variants (Baseline I, Baseline II, and the Hierarchical Pipeline) were evaluated under the same expanding-window cross-validation protocol for imputation quality and the same fixed temporal split (training: 2015–2024, test: January–March 2025) for forecasting performance, using identical model configurations across both AutoML frameworks.
- (b) Did you specify all the necessary details of your evaluation (e.g., data splits, preprocessing, search spaces, hyperparameter tuning details and results, etc.)? [Yes] The Experimental Setup section specifies the data splits, preprocessing variants, model configurations (DeepAR: context window 168 h, 2 layers, hidden size 64, 50 epochs; XGBoost: 300 estimators, learning rate 0.05; SARIMA: auto-selected orders; Prophet: default structure), and evaluation metrics (MASE scaled to a 24-hour seasonal period).
- (c) Did you repeat your experiments (e.g., across multiple random seeds or splits) to account for the impact of randomness in your methods or data? [No] The imputation evaluation uses an expanding-window cross-validation with 575 fold-station-gap combinations, providing robustness against temporal variability across the ten-year evaluation period. The forecasting evaluation uses a single fixed temporal split, which is standard practice for time series to avoid look-ahead bias; stochastic models (DeepAR) were trained once without multiple seed repetitions, which is acknowledged as a limitation.
- (d) Did you report the uncertainty of your results (e.g., the standard error across random seeds or splits)? [Yes] Standard deviations are reported for all imputation quality metrics across the 575 tsCV fold-station-gap combinations (Table 4, appendix). Forecasting MASE values are reported as point estimates per station and model, consistent with the fixed temporal split design.
- (e) Did you report the statistical significance of your results? [Yes]
- (f) Did you use enough repetitions, datasets, and/or benchmarks to support your claims? [Yes] The evaluation spans six stations, ten years of hourly measurements, two AutoML frameworks, four forecasting models, three preprocessing variants, and 575 tsCV fold-station-gap combinations for imputation quality. The consistency of results across metrics, gap sizes, and folds provides robustness to support the claims made.
- (g) Did you compare performance over time and describe how you selected the maximum runtime? [No] Runtime was not the primary focus of the study. Fit and prediction wall-clock times are logged in the repository outputs but are not analysed in the paper.

- (h) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? **[Yes]** Computational requirements are documented in the repository README at <https://github.com/dallasta-gui/Preprocessing-for-AutoML>, including per-notebook CPU/GPU requirements, minimum and recommended RAM, and estimated wall-clock times. All experiments were run on local hardware without external cluster or cloud resources.
- (i) Did you run ablation studies to assess the impact of different components of your approach? **[Yes]** The three-variant comparison (Baseline I, Baseline II, Hierarchical Pipeline) functions as an ablation study, isolating the contribution of the hierarchical gap-filling strategy relative to naive baselines and assessing how preprocessing quality affects downstream AutoML forecasting behaviour.
3. With respect to the code used to obtain your results...
- (a) Did you include the code, data, and instructions needed to reproduce the main experimental results, including all dependencies (e.g., requirements.txt with explicit versions), random seeds, an instructive README with installation instructions, and execution commands (either in the supplemental material or as a URL)? **[Yes]** The repository at <https://github.com/dallasta-gui/Preprocessing-for-AutoML> contains the three notebooks (DO_gap_filling_pipeline.ipynb, sktime.ipynb, AutoGluonTS.ipynb), all six raw CSV data files, a detailed README with execution instructions, and a dependency table with explicit versions (Python 3.10, sktime ≥ 0.30 , AutoGluon-TimeSeries ≥ 1.2 , prophet 1.3, statsmodels ≥ 0.14 , neuralforecast ≥ 1.7).
- (b) Did you include a minimal example to replicate results on a small subset of the experiments or on toy data? **[No]** No minimal standalone example is provided. The notebooks are self-contained and can be run on individual stations; EF07, with the lowest rate of missing values (6.51%), represents the simplest case for initial verification.
- (c) Did you ensure sufficient code quality and documentation so that someone else can execute and understand your code? **[Yes]** Each notebook is self-documented, and the README describes every pipeline stage, the format of input data, and the outputs generated by each notebook, enabling independent reproduction of the results.
- (d) Did you include the raw results of running your experiments with the given code, data, and instructions? **[No]** The raw output CSV files (e.g., sktime_metrics_FINAL_completa.csv, AutoGluonTS_metrics_FINAL_completa.csv) are not included in the repository. Only the raw input data are provided; output files are generated by running the notebooks.
- (e) Did you include the code, additional data, and instructions needed to generate the figures and tables in your paper based on the raw results? **[Yes]** All figures and metric tables reported in the paper are generated directly by the notebooks, including imputation bar plots, per-station forecast plots, and the station map. The README documents the expected output files and their correspondence to the paper figures.
4. If you used existing assets (e.g., code, data, models)...
- (a) Did you cite the creators of used assets? **[Yes]** AutoGluon-TimeSeries, sktime, Prophet, and the CETESB SIMQUA monitoring network are all cited in the paper.
- (b) Did you discuss whether and how consent was obtained from people whose data you're using/curating if the license requires it? **[Yes]** The dissolved oxygen data are publicly available through CETESB's SIMQUA platform, which provides open access to environmental monitoring records from the State of São Paulo without restrictions on research use.

- (c) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [Yes] The dataset consists exclusively of automated sensor measurements of dissolved oxygen concentrations from river monitoring stations. It contains no personally identifiable information or offensive content.
5. If you created/released new assets (e.g., code, data, models)...
- (a) Did you mention the license of the new assets (e.g., as part of your code submission)? [Yes] The repository is released under the MIT License, as specified in the LICENSE file at <https://github.com/dallasta-gui/Preprocessing-for-AutoML>.
 - (b) Did you include the new assets either in the supplemental material or as a URL (to, e.g., GitHub or Hugging Face)? [Yes] The code and data are publicly available at <https://github.com/dallasta-gui/Preprocessing-for-AutoML>.
6. If you used crowdsourcing or conducted research with human subjects...
- (a) Did you include the full text of instructions given to participants and screenshots, if applicable? [N/A] This work does not involve crowdsourcing or human subjects.
 - (b) Did you describe any potential participant risks, with links to institutional review board (IRB) approvals, if applicable? [N/A]
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [N/A]
7. If you included theoretical results...
- (a) Did you state the full set of assumptions of all theoretical results? [N/A] This paper does not include theoretical results.
 - (b) Did you include complete proofs of all theoretical results? [N/A]

A Regime-specific quality-control and gap-filling parameters

Table 6: Regime-specific quality-control and gap-filling parameters, defined in consultation with a domain expert. A window is flagged as a constancy artefact only when the observed range, standard deviation, and absolute regression slope all fall simultaneously below the thresholds listed below (Section 4.1). *min_obs* is the minimum number of non-missing hourly observations required within the rolling window for the window to be evaluated. *max_small_gap* denotes the largest gap length (in hours) treated as “short” for direct linear interpolation in the hierarchical gap-filling strategy (Section 4.2). Only the Intermediate, Diel-productive, and Hypoxic-impacted regimes were represented among the six evaluated stations; the Regulated/damped and Turbulent/headwater regimes, while supported by the classification framework, did not occur in this dataset.

Regime	<i>window</i>	<i>eps</i> (mg/L)	<i>std_max</i> (mg/L)	<i>slope_max</i> (mg/L/h)	<i>min_obs</i>	<i>max_small_gap</i> (h)
Hypoxic-impacted	72 h	0.30	0.15	0.010	48	12
Regulated/damped	72 h	0.25	0.12	0.008	48	8
Diel-productive	48 h	0.20	0.10	0.006	36	6
Turbulent/headwater	24 h	0.15	0.08	0.005	24	4
Intermediate	48 h	0.25	0.12	0.008	36	6

B Visual Results of Data Quality Control

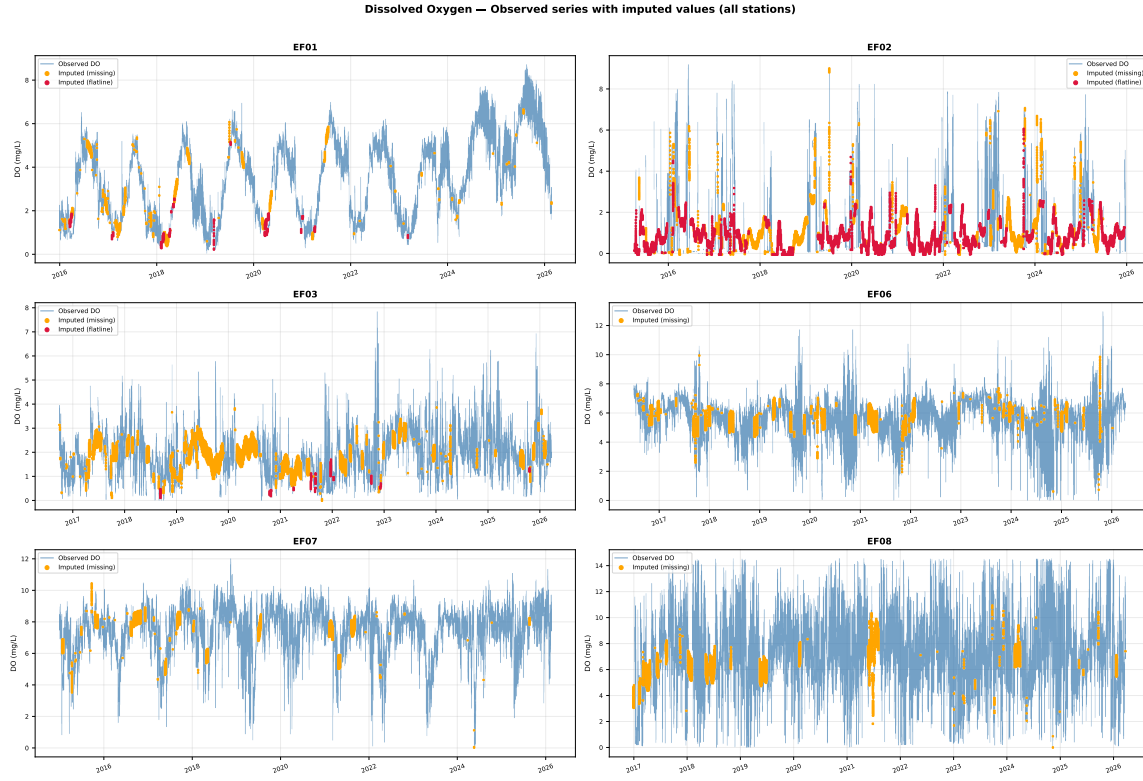


Figure 6: Dissolved oxygen time series for all six monitoring stations after quality control and hierarchical gap-filling. Blue lines represent observed values retained after quality control. Orange points indicate positions reconstructed due to originally missing observations. Red points indicate positions where constancy detection flagged extended flat-line segments as probable sensor artefacts; these observations were removed during quality control and subsequently reconstructed by the imputation pipeline. The volume and temporal distribution of red points reflects the severity of sensor fouling or drift at each station: EF02 exhibits the densest concentration of constancy artefacts, consistent with its hypoxic-impacted regime and chronic near-zero readings that are structurally indistinguishable from sensor failure. EF06 and EF07 show no constancy artefacts, indicating that their diel and intermediate dynamics were never suppressed to flat-line behaviour over the evaluation period.

C Detailed Experimental Results

Table 7: Time series cross-validation summary ($N = 575$ combinations). Lower is better for DTW, |Bias|, KS, MAED, CETrend; higher for DA, MDA.

Method	DTW	Bias	KS	DA	MDA	MAED	CETrend
Baseline I	4.33 ± 5.6	-0.08 ± 0.8	0.74 ± 0.24	0.00 ± 0.0	0.14 ± 0.28	0.12 ± 0.13	2.02 ± 0.7
Baseline II	2.90 ± 3.7	-0.14 ± 0.4	0.52 ± 0.23	0.60 ± 0.19	0.57 ± 0.21	0.12 ± 0.12	2.16 ± 1.7
Hierarchical	2.45 ± 3.3	-0.10 ± 0.4	0.46 ± 0.22	0.68 ± 0.18	0.62 ± 0.21	0.12 ± 0.13	1.66 ± 2.0

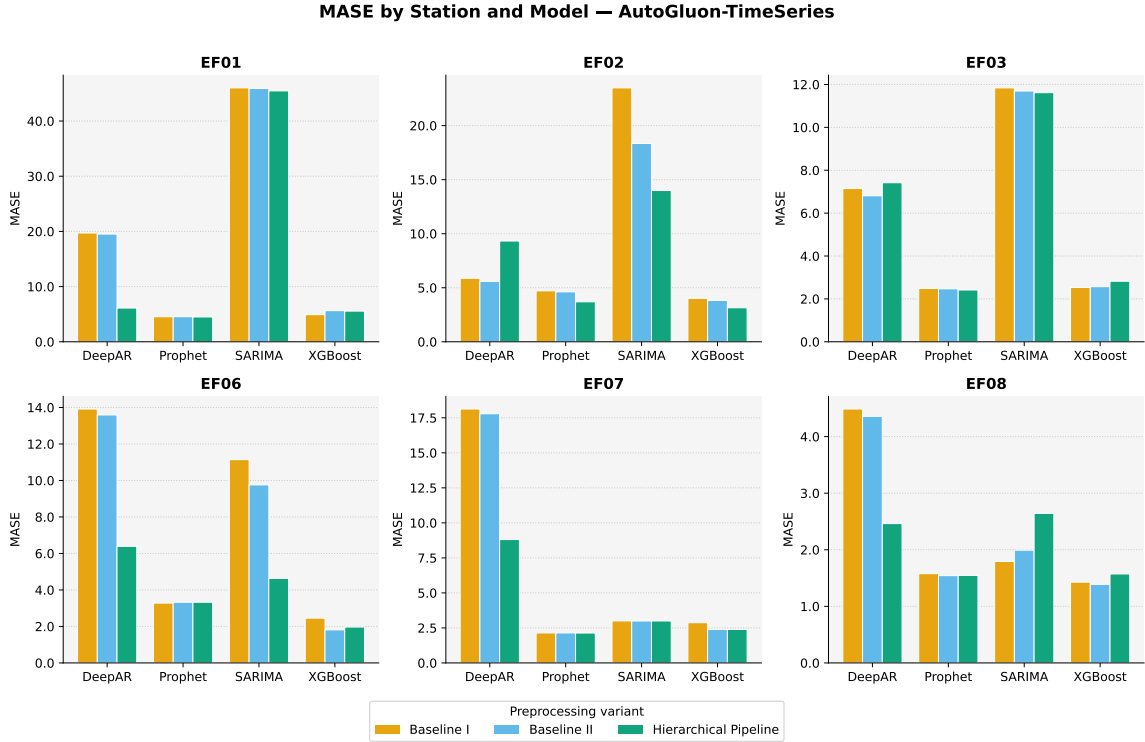


Figure 7: MASE per station and forecasting model under AutoGluon-TimeSeries, for the three preprocessing variants (Baseline I, Baseline II, and the Hierarchical Pipeline). Lower values indicate better forecast accuracy. Vertical axes are not shared across panels, as error magnitudes differ substantially between stations.

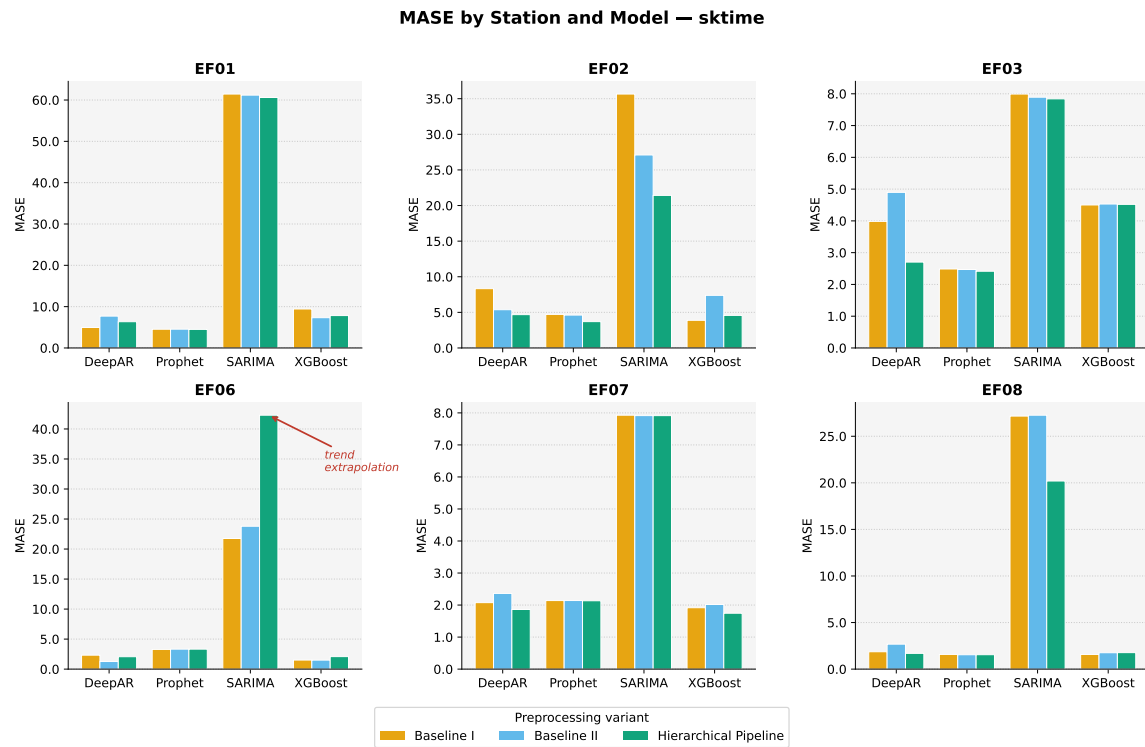


Figure 8: MASE per station and forecasting model under sktime, for the same three preprocessing variants as Figure 7. Lower values indicate better forecast accuracy; vertical axes are not shared across panels. The annotated bar for SARIMA at EF06 under the Hierarchical Pipeline corresponds to the trend extrapolation pathology, in which the differencing scheme extrapolates a persistent linear drift over the 91-day horizon.

Table 8: MASE per station and model — AutoGluon-TimeSeries. Δ_I and Δ_{II} : percentage change from Baseline I and Baseline II to the Hierarchical Pipeline, respectively. *Signal* indicates whether the Hierarchical Pipeline improved ($\downarrow$) or degraded ($\uparrow$) MASE for that model, with the direction determined by whichever of Δ_I or Δ_{II} has the larger absolute magnitude.

Station	Model	BI	BII	Hierarchical	Δ_I	Δ_{II}	Signal
EF01	DeepAR	19.71	19.52	6.11	-68.98%	-68.70%	$\downarrow$
	Prophet	4.54	4.55	4.49	-1.10%	-1.32%	$\downarrow$
	SARIMA	46.01	45.89	45.46	-1.19%	-0.94%	$\downarrow$
	XGBoost	4.91	5.64	5.56	+13.24%	-1.42%	$\uparrow$
EF02	DeepAR	5.87	5.59	9.32	+58.77%	+66.73%	$\uparrow$
	Prophet	4.71	4.62	3.70	-21.44%	-19.91%	$\downarrow$
	SARIMA	23.49	18.35	13.99	-40.49%	-23.76%	$\downarrow$
	XGBoost	4.02	3.83	3.15	-21.64%	-17.76%	$\downarrow$
EF03	DeepAR	7.15	6.81	7.42	+3.78%	+8.96%	$\uparrow$
	Prophet	2.49	2.47	2.41	-3.21%	-2.43%	$\downarrow$
	SARIMA	11.84	11.69	11.62	-1.86%	-0.60%	$\downarrow$
	XGBoost	2.53	2.57	2.82	+11.46%	+9.73%	$\uparrow$
EF06	DeepAR	13.92	13.59	6.39	-54.10%	-52.98%	$\downarrow$
	Prophet	3.28	3.33	3.33	+1.52%	+0.00%	$\uparrow$
	SARIMA	11.14	9.76	4.63	-58.44%	-52.56%	$\downarrow$
	XGBoost	2.46	1.81	1.97	-19.92%	+8.84%	$\downarrow$
EF07	DeepAR	18.13	17.79	8.81	-51.40%	-50.48%	$\downarrow$
	Prophet	2.14	2.14	2.13	-0.47%	-0.47%	$\downarrow$
	SARIMA	3.00	3.00	3.00	-0.03%	-0.03%	$\downarrow$
	XGBoost	2.87	2.40	2.39	-16.72%	-0.42%	$\downarrow$
EF08	DeepAR	4.49	4.36	2.46	-45.21%	-43.58%	$\downarrow$
	Prophet	1.58	1.54	1.55	-1.90%	+0.65%	$\downarrow$
	SARIMA	1.79	1.99	2.64	+47.49%	+32.66%	$\uparrow$
	XGBoost	1.43	1.39	1.57	+9.79%	+12.95%	$\uparrow$

Table 9: MASE per station and model — sktime. Δ_I and Δ_{II} : percentage change from Baseline I and Baseline II to the Hierarchical Pipeline, respectively. *Signal* indicates whether the Hierarchical Pipeline improved ($\downarrow$) or degraded ($\uparrow$) MASE for that model, with the direction determined by whichever of Δ_I or Δ_{II} has the larger absolute magnitude.

Station	Model	BI	BII	Hierarchical	Δ_I	Δ_{II}	Signal
EF01	DeepAR	4.95	7.71	6.35	+28.40%	-17.66%	$\uparrow$
	Prophet	4.54	4.55	4.49	-1.10%	-1.32%	$\downarrow$
	SARIMA	61.44	61.18	60.61	-1.35%	-0.93%	$\downarrow$
	XGBoost	9.43	7.32	7.84	-16.87%	+7.10%	$\downarrow$
EF02	DeepAR	8.33	5.37	4.68	-43.82%	-12.85%	$\downarrow$
	Prophet	4.71	4.62	3.70	-21.44%	-19.91%	$\downarrow$
	SARIMA	35.65	27.10	21.42	-39.89%	-20.96%	$\downarrow$
	XGBoost	3.88	7.39	4.57	+17.79%	-38.16%	$\uparrow$
EF03	DeepAR	3.98	4.90	2.70	-32.14%	-44.83%	$\downarrow$
	Prophet	2.49	2.47	2.41	-3.21%	-2.43%	$\downarrow$
	SARIMA	7.99	7.89	7.84	-1.90%	-0.66%	$\downarrow$
	XGBoost	4.50	4.53	4.52	+0.41%	-0.24%	$\uparrow$
EF06	DeepAR	2.32	1.26	2.07	-10.97%	+63.37%	$\downarrow$
	Prophet	3.28	3.33	3.33	+1.52%	+0.02%	$\uparrow$
	SARIMA	21.76	23.80	42.29	+94.37%	+77.69%	$\uparrow$
	XGBoost	1.51	1.50	2.08	+37.97%	+38.84%	$\uparrow$
EF07	DeepAR	2.08	2.36	1.86	-10.28%	-21.22%	$\downarrow$
	Prophet	2.14	2.14	2.13	-0.47%	-0.47%	$\downarrow$
	SARIMA	7.93	7.92	7.91	-0.17%	-0.01%	$\downarrow$
	XGBoost	1.92	2.02	1.74	-9.14%	-13.49%	$\downarrow$
EF08	DeepAR	1.86	2.68	1.69	-9.33%	-36.92%	$\downarrow$
	Prophet	1.58	1.54	1.55	-1.90%	+0.65%	$\downarrow$
	SARIMA	27.17	27.26	20.19	-25.71%	-25.91%	$\downarrow$
	XGBoost	1.58	1.76	1.77	+12.03%	+0.59%	$\uparrow$

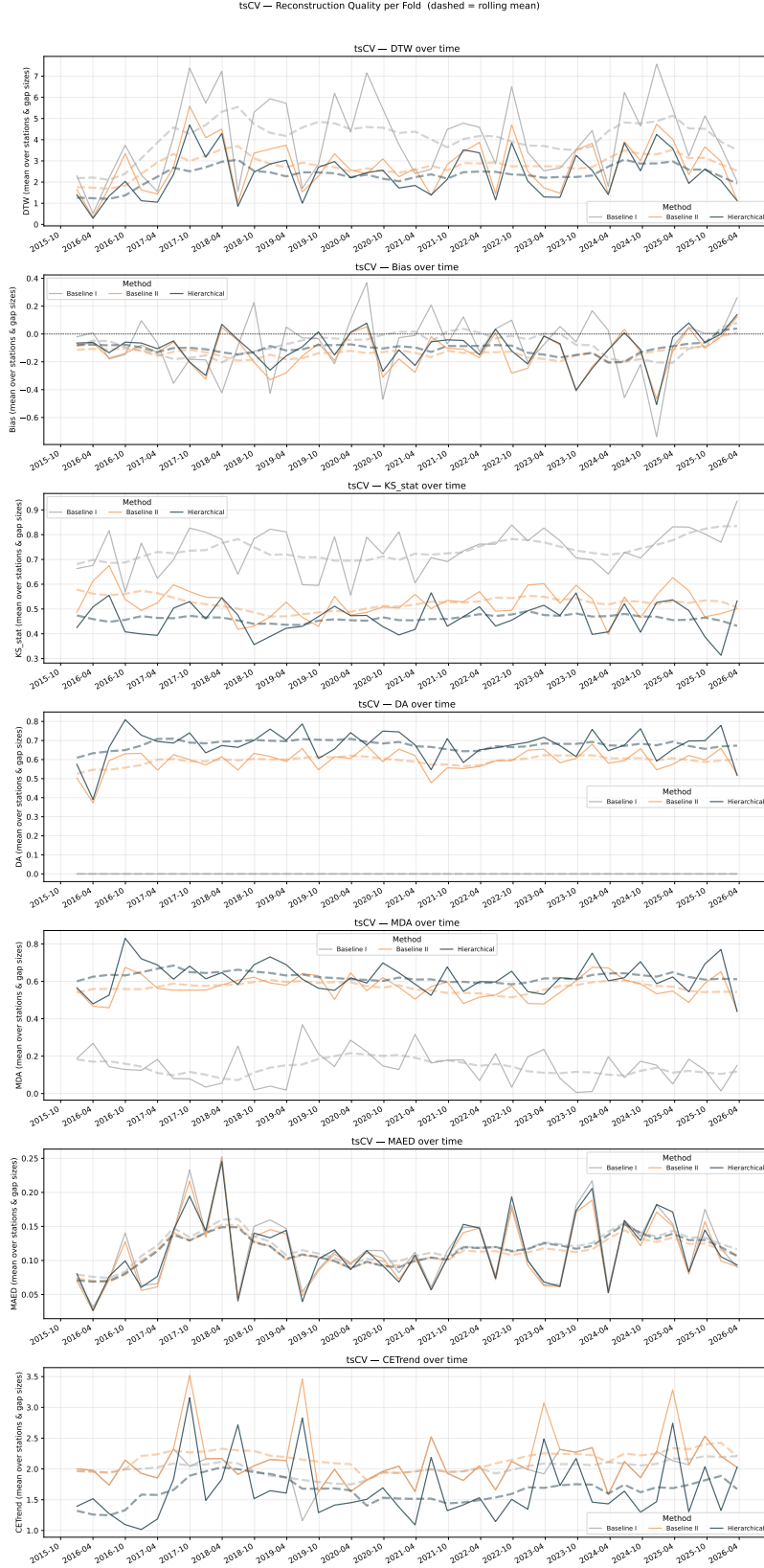
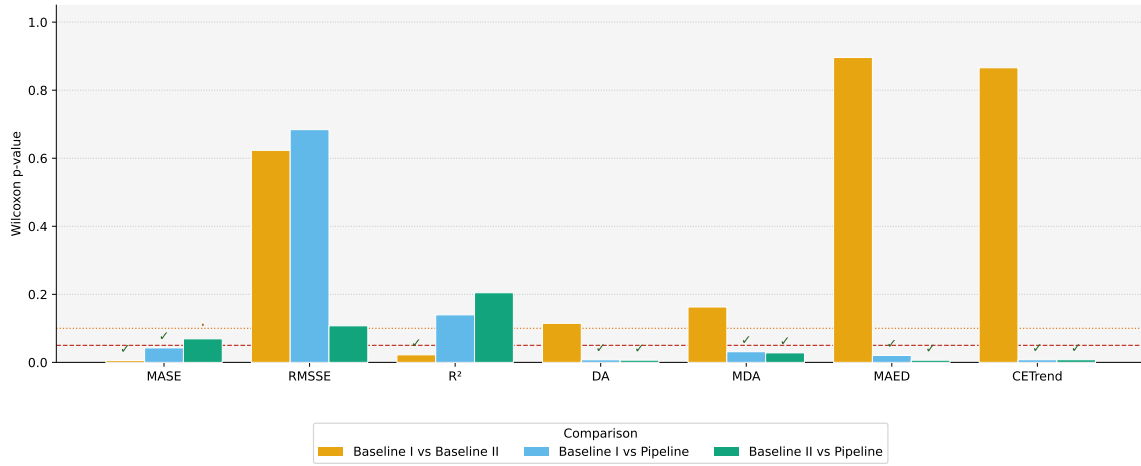
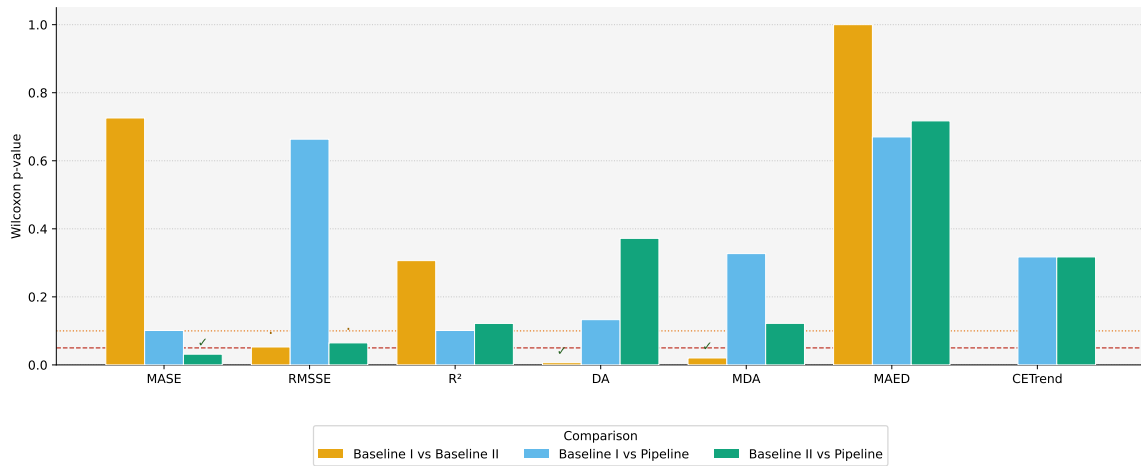


Figure 9: Temporal evolution of reconstruction metrics across tsCV folds. Dashed lines: rolling means. The Hierarchical Pipeline maintains consistently lower DTW and KS statistics with no degradation over time.

D Results of Statistical Tests

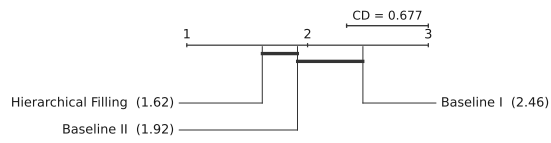


(a) AutoGluon-TimeSeries. The Hierarchical Pipeline differs significantly from both baselines in DA, MDA, MAED, and CETrend, while the two baselines differ from each other mainly in MASE and R^2 .

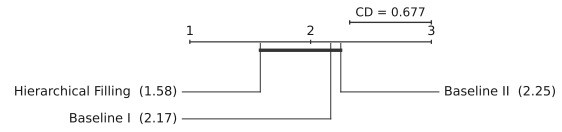


(b) sktime. No bar is shown for Baseline I vs. Baseline II in CETrend, as paired differences were identically zero across all blocks. Unlike AutoGluon-TimeSeries, significant differences here concentrate between the two baselines themselves (DA, MDA) rather than between either baseline and the Pipeline.

Figure 10: Wilcoxon signed-rank test for the pairwise comparison of Baseline I, Baseline II, and the Hierarchical Pipeline. Bars show the two-sided Wilcoxon p -value for each pairwise comparison, computed per metric across all Model $\times$ Station blocks (4 models $\times$ 6 stations = 24 paired observations). Dashed red line: $p = 0.05$; dotted orange line: $p = 0.10$. ✓ marks $p < 0.05$; · marks $p < 0.10$.



(a) AutoGluon-TimeSeries ($n = 24$ model-station pairs). Pipelines joined by a bold bar do not differ significantly at $\alpha = 0.05$ (Nemenyi post-hoc).



(b) sktime ($n = 24$ model-station pairs). Pipelines joined by a bold bar do not differ significantly at $\alpha = 0.05$ (Nemenyi post-hoc).

Figure 11: Critical-difference diagrams comparing Baseline I, Baseline II, and the Hierarchical Pipeline across all model-station pairs, based on the Friedman/Nemenyi test at $\alpha = 0.05$.